

HE2003 Econometrics I

Tutorial 3 (Week 4) – Problems & Solutions

Academic Year 2026/2027, Semester 1

Quantitative Research Society @NTU

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Overview

- **Lecture coverage:** Lecture 3 — the true-vs-estimated comparison (population objects vs. OLS objects), the algebraic properties of OLS residuals that follow from the first-order conditions, and the Least Squares Assumptions (with Assumption 1, zero conditional mean, as the conceptual centrepiece).
- **Questions:** 4 questions from Tutorial Question Set No. 3.
- **Notation:** as in lectures — u_i population error, $\hat{u}_i$ OLS residual; β_0, β_1 population (unknown) parameters, $\hat{\beta}_0, \hat{\beta}_1$ OLS estimators; “Least Squares Assumption 1/2/3” (never SLR.1–5).

Question 1 (Population vs. OLS: five core objects)

Problem

Explain the following: population regression line, OLS regression line, error term, OLS residual, and OLS predicted value. (*Tutorial Question Set No. 3, Q1.*)

Solution

Method 1: Definitions

Model: $Y_i = \beta_0 + \beta_1 X_i + u_i$, $i = 1, \dots, n$.

- Population regression line: $\beta_0 + \beta_1 X$ – unknown, fixed; $= E[Y_i | X_i]$ under Least Squares Assumption 1.
- Error term: $u_i = Y_i - (\beta_0 + \beta_1 X_i)$ – all other factors affecting Y_i ; unobserved.
- OLS regression line: $\hat{\beta}_0 + \hat{\beta}_1 X$, $\hat{\beta}_0, \hat{\beta}_1$ minimising $\sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2$ – computed from the sample.
- OLS predicted value: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ – height of the OLS line at X_i .
- OLS residual: $\hat{u}_i = Y_i - \hat{Y}_i$ – computable counterpart of u_i ; $\hat{u}_i \neq u_i$ in general.

Population: $\beta_0 + \beta_1 X_i$, $u_i = Y_i - (\beta_0 + \beta_1 X_i)$ (unobserved)
Sample: $\hat{\beta}_0 + \hat{\beta}_1 X_i$, $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$, $\hat{u}_i = Y_i - \hat{Y}_i$ (all computable)

Question 2 (Residual properties from the FOCs)

Problem

- (a) Show that $\sum_{i=1}^n \hat{u}_i = 0$ by using the first-order condition (FOC) of the OLS estimators.
- (b) Show that $\sum_{i=1}^n \hat{u}_i X_i = 0$ by using the FOC of the OLS estimators.
- (c) Show that $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$. [Hint: $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = \sum_{i=1}^n \hat{u}_i (\hat{\beta}_0 + \hat{\beta}_1 X_i)$.]

(Tutorial Question Set No. 3, Q2.)

Solution

Method 1: First-order conditions

FOCs of $\min_{b_0, b_1} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2$ at $(\hat{\beta}_0, \hat{\beta}_1)$, with $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$:

$$-2 \sum_{i=1}^n \hat{u}_i = 0 \quad (\text{FOC } b_0)$$

$$-2 \sum_{i=1}^n \hat{u}_i X_i = 0 \quad (\text{FOC } b_1)$$

(a)

$$(\text{FOC } b_0) \implies \boxed{\sum_{i=1}^n \hat{u}_i = 0.}$$

(b)

$$(\text{FOC } b_1) \implies \boxed{\sum_{i=1}^n \hat{u}_i X_i = 0.}$$

(c)

$$\begin{aligned} \sum_{i=1}^n \hat{u}_i \hat{Y}_i &= \hat{\beta}_0 \sum_{i=1}^n \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^n \hat{u}_i X_i & (\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i) \\ &= \hat{\beta}_0(0) + \hat{\beta}_1(0) & ((a), (b)) \\ \implies &\boxed{\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0.} \end{aligned}$$

Method 2: closed-form estimators

$$\begin{aligned}
\hat{u}_i &= (Y_i - \bar{Y}) - \hat{\beta}_1(X_i - \bar{X}) && (\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X}) \\
\sum_{i=1}^n \hat{u}_i &= \sum_{i=1}^n (Y_i - \bar{Y}) - \hat{\beta}_1 \sum_{i=1}^n (X_i - \bar{X}) = 0 && (\text{deviations sum to 0}) \\
\sum_{i=1}^n \hat{u}_i X_i &= \sum_{i=1}^n \hat{u}_i (X_i - \bar{X}) + \bar{X} \sum_{i=1}^n \hat{u}_i \\
&= \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) - \hat{\beta}_1 \sum_{i=1}^n (X_i - \bar{X})^2 + 0 && (\hat{u}_i, \sum_{i=1}^n \hat{u}_i = 0 \text{ above}) \\
&= 0 && (\text{def. of } \hat{\beta}_1) \\
\Rightarrow \sum_{i=1}^n \hat{u}_i \hat{Y}_i &= \hat{\beta}_0 \sum_{i=1}^n \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^n \hat{u}_i X_i = 0 && (\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i)
\end{aligned}$$

$\sum_{i=1}^n \hat{u}_i = 0,$	$\sum_{i=1}^n \hat{u}_i X_i = 0,$	$\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0.$
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Question 3 (Assumption 1 by diagrams, and a counterexample)

Problem

- (a) Explain the meaning of the OLS Assumption 1 by using diagrams.
- (b) Explain how we can obtain the figure shown on slide No. 11 for Lecture 3.
- (c) Give an example where $E[u] = 0$ but $E[u | X] \neq 0$ by using diagrams.

(Tutorial Question Set No. 3, Q3.)

Solution

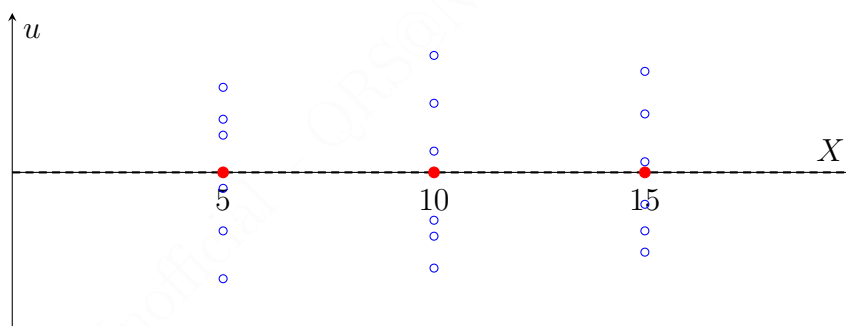
Method 1: Figures

(a)

$$E[u_i | X_i] = 0 : \quad E[u_i | X_i = 5] = E[u_i | X_i = 10] = E[u_i | X_i = 15] = \dots = 0$$

At every X the errors lie above and below 0 and average to 0 (red dots, Figure 1).

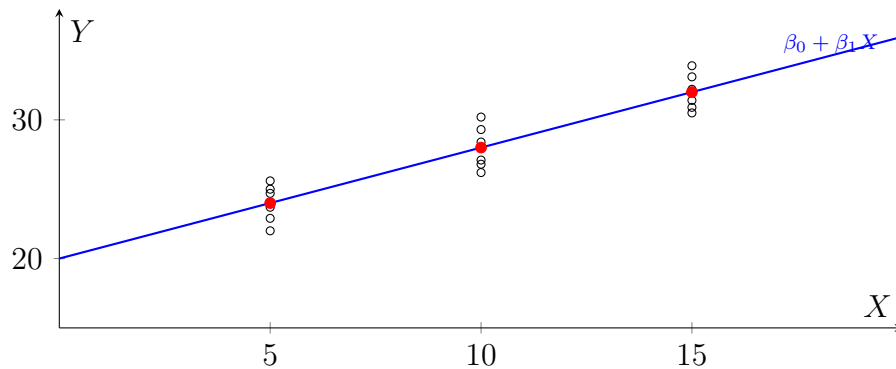
Figure 1: u_i vs. X_i – zero conditional mean at $X = 5, 10, 15$



(b)

$$\begin{aligned}
 Y_i &= \beta_0 + \beta_1 X_i + u_i \\
 E[Y_i | X_i] &= \beta_0 + \beta_1 X_i + E[u_i | X_i] && \text{(linearity; } X_i \text{ given)} \\
 &= \beta_0 + \beta_1 X_i && \text{(Least Squares Assumption 1)}
 \end{aligned}$$

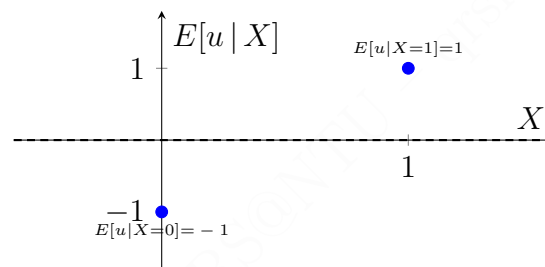
Figure 2 = Figure 1 with each point lifted by $\beta_0 + \beta_1 X_i$; the conditional means (red) lie on the line.

Figure 2: Y_i vs. X_i – data centred on the population line at $X = 5, 10, 15$ 

(c) $X \in \{0, 1\}$, $\Pr(X = 0) = \Pr(X = 1) = \frac{1}{2}$, $E[u | X = 0] = -1$, $E[u | X = 1] = 1$:

$$E[u] = \frac{1}{2}(-1) + \frac{1}{2}(1) = 0 \quad (\text{Law of Total Expectation})$$

$$E[u | X = 0] = -1 \neq 0, \quad E[u | X = 1] = 1 \neq 0$$

Figure 3: binary X – $E[u] = 0$ but $E[u | X] \neq 0$ 

Second counterexample: continuous regressor

$X \sim N(0, 1)$, $u = X^2 - 1$:

$$E[u] = \text{Var}(X) + (E[X])^2 - 1 = 1 - 1 = 0$$

$$E[u | X = x] = x^2 - 1 \neq 0 \quad (\text{zero only at } x = \pm 1)$$

Assumption 1: $E[u_i | X_i] = 0$ for every value of $X_i \implies E[Y_i | X_i] = \beta_0 + \beta_1 X_i$.

Counterexamples: binary X with $E[u|X=0]=-1$, $E[u|X=1]=1$;
continuous $u = X^2 - 1$, $X \sim N(0, 1)$ – both have $E[u] = 0$ but $E[u|X] \neq 0$.

Question 4 (Binary regressor: SSS vs. SOH wages)

Problem

Suppose we want to find out if there exists a difference in average wages between graduates from the School of Social Sciences (SSS) and graduates from the School of Humanities (SOH), using the linear regression model with an indicator variable ($X = 1$ or 0).

- Write down the linear regression model for this problem. Define the dependent variable, explanatory variable, and error term in your model.
- Write down the corresponding OLS Assumption 1. Explain in what case the OLS Assumption 1 is likely to hold/unlikely to hold.
- Now suppose that the OLS Assumption 1 does hold; explain the specific meaning of the parameters β_0 and β_1 using conditional expectation. [Hint: derive the conditional expectation $E[Y_i | X_i]$ for $X_i = 1$ and $X_i = 0$, respectively.]
- Illustrate the meaning of β_0 and β_1 by using a diagram.

(Tutorial Question Set No. 3, Q4.)

Solution

Method 1: Conditional expectation

(a)

$$\text{wage}_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n$$

- wage_i : wage of graduate i (dependent variable);
- $X_i = 1$ if SSS, $X_i = 0$ if SOH (explanatory variable);
- u_i : other wage factors – ability, family background, experience, effort (error term).

(b)

$$E[u_i | X_i] = 0 \iff E[u_i | X_i = 1] = 0 \text{ and } E[u_i | X_i = 0] = 0$$

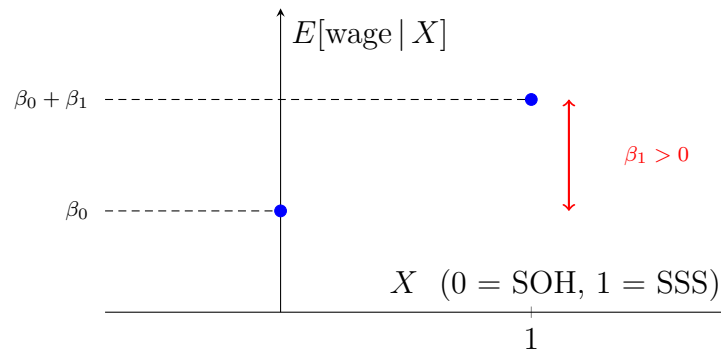
- Likely to hold: random assignment to SSS/SOH, so the groups are balanced on ability.
- Unlikely to hold: self-selection by ability, interests or background, correlated with X_i and wage.

(c)

$$\begin{aligned} E[Y_i | X_i] &= \beta_0 + \beta_1 X_i + E[u_i | X_i] = \beta_0 + \beta_1 X_i \quad (\text{Least Squares Assumption 1}) \\ E[Y_i | X_i = 0] &= \beta_0, \quad E[Y_i | X_i = 1] = \beta_0 + \beta_1 \\ \implies \beta_1 &= E[Y_i | X_i = 1] - E[Y_i | X_i = 0] \end{aligned}$$

- β_0 : mean wage of SOH graduates;
- β_1 : difference in mean wages, SSS minus SOH – a level shift, not a slope.

(d) β_0 at $X = 0$, $\beta_0 + \beta_1$ at $X = 1$ (case $\beta_1 > 0$).



Method 2: orthogonality of residuals

n_1, n_0 : numbers with $X_i = 1, 0$; $\bar{Y}_1, \bar{Y}_0$: sample mean wage of SSS, SOH.

$$\sum_{X_i=1} \hat{u}_i = \sum_{i=1}^n \hat{u}_i X_i = 0 \implies n_1(\bar{Y}_1 - \hat{\beta}_0 - \hat{\beta}_1) = 0 \quad (\text{Q2(b)})$$

$$\sum_{X_i=0} \hat{u}_i = \sum_{i=1}^n \hat{u}_i - \sum_{i=1}^n \hat{u}_i X_i = 0 \implies n_0(\bar{Y}_0 - \hat{\beta}_0) = 0 \quad (\text{Q2(a), (b)})$$

$$\therefore \hat{\beta}_0 = \bar{Y}_0, \quad \hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$$

$$\text{wage}_i = \beta_0 + \beta_1 X_i + u_i, \quad X_i = 1 \text{ (SSS)}, 0 \text{ (SOH)}$$

Assumption 1: $E[u_i|X_i=1] = E[u_i|X_i=0] = 0$; holds under random assignment,
fails under self-selection by ability/background

$$\beta_0 = E[\text{wage}|\text{SOH}], \quad \beta_1 = E[\text{wage}|\text{SSS}] - E[\text{wage}|\text{SOH}] \text{ (not a slope)}$$

$$\text{Sample: } \hat{\beta}_0 = \bar{Y}_0, \quad \hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$$

Final Answers

1. Population line $\beta_0 + \beta_1 X$; error $u_i = Y_i - (\beta_0 + \beta_1 X_i)$ (both unobserved); OLS line $\hat{\beta}_0 + \hat{\beta}_1 X$; predicted value $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$; residual $\hat{u}_i = Y_i - \hat{Y}_i$ (all computed).
2. $\sum_{i=1}^n \hat{u}_i = 0$; $\sum_{i=1}^n \hat{u}_i X_i = 0$; $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$.
3. (a) $E[u_i|X_i] = 0$ at every X (Figure 1). (b) $E[Y_i|X_i] = \beta_0 + \beta_1 X_i$ (Figure 2). (c) Binary X : $E[u|X=0] = -1$, $E[u|X=1] = 1$ (Figure 3); continuous: $u = X^2 - 1$, $X \sim N(0, 1)$; both have $E[u] = 0$, $E[u|X] \neq 0$.
4. (a) $\text{wage}_i = \beta_0 + \beta_1 X_i + u_i$, $X_i = 1$ (SSS) or 0 (SOH). (b) $E[u_i|X_i=1] = E[u_i|X_i=0] = 0$; holds under random assignment, fails under self-selection. (c) $\beta_0 = E[\text{wage}|\text{SOH}]$, $\beta_1 = E[\text{wage}|\text{SSS}] - E[\text{wage}|\text{SOH}]$. (d) β_0 at $X = 0$, $\beta_0 + \beta_1$ at $X = 1$. Sample: $\hat{\beta}_0 = \bar{Y}_0$, $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$.